Financial Time Series Analysis

Lecture 6

- TA Session Friday April 9, 2021, 11:00am - 12:00pm (EDT)

Outline

- Parameter Estimation
  - Least Squares
  - Yule Walker Equations
  - Likelihood Methods
- Statsmodels fitting ARMA Models
- Forecasting

Parameter Estimation: 1

- We begin with the estimation of autoregressive models. If we have assumptions about the distribution of ϵ_t, then we can use standard maximum likelihood methods. On the other hand, if the error process is only assumed to be $\mathrm{WN}(0, \sigma^2)$, then we need to fall back on moment methods.
- We want to assume we are working with a weakly stationary model
- The standard autoregressive model for Y_t with $E(Y_t) = 0$ can be written in vector form as

$$Y_t = \mathbf{Y}_{t-1}^T \boldsymbol{\phi} + \epsilon_t,$$

where $\mathbf{Y}_{t-1} = (Y_{t-1}, \cdots, Y_{t-p})^T$ and $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)^T$.

Parameter Estimation: 2

- Putting this together with data $(Y_1, \cdots, Y_n)$, we can find the least squares solution for ϕ:

$$\hat{\phi} = (\sum_{t=p+1}^{n} \mathbf{Y}_{t-1}\mathbf{Y}_{t-1}^T)^{-1}(\sum_{t=p+1}^{n} \mathbf{Y}_{t-1}Y_t).$$

- If the ϵ_t sequence is iid $N(0, \sigma^2)$, then the estimator is also the MLE.
- In the simple case that $p = 1$, the formula simplifies to

$$\hat{\phi} = \sum_{t=1}^{n-1} Y_t Y_{t+1} / \sum_{t=1}^{n-1} Y_t^2.$$

Parameter Estimation: 3

- The least squares estimators do not make any assumptions about the error distribution, yet given that the errors are white noise, we can work out the asymptotic distribution of those estimators. In particular they are asymptotically normal:

  $$\sqrt{n}(\hat{\phi} - \phi) \Rightarrow N_p(\mathbf{0}, \sigma_\epsilon^2 \mathbf{\Gamma}_p^{-1}),$$

  where

  $$\mathbf{\Gamma}_p = E(\mathbf{Y}^T\mathbf{Y}),$$

  with $\mathbf{Y} = (Y_1, \cdots, Y_p)$.
- The (ij) element of this matrix is $E(Y_iY_j) = \gamma(i-j)$.

Parameter Estimation: 4

- For p=1, the AR(1) model, we have $\mathbf{\Gamma}_1 = (\gamma(0)) = \sigma_\epsilon^2/(1-\phi_1^2)$.
- This gives

$$\sqrt{n}(\hat{\phi}_1 - \phi_1) \Rightarrow N(0, \sigma_\epsilon^2/\sigma_\epsilon^2/(1-\phi_1^2)) = N(0, (1-\phi_1^2)).$$

- Approximate confidence intervals can be generated and hypothesis tests conducted about the parameter ϕ_1 or the parameters ϕ using methods based on the normal distribution.
- We next turn to the familiar Yule-Walker (YW) equations approach.

Yule Walker Equations: 1

- From Homework #2 students are familiar with the YW equations.
- Multiply the standard autoregressive equation, $\phi(B)Y_t = \epsilon_t$ by Y_{t-k} and take expectations. This leads to a set of linear equations for the autocovariance values, and if one normalizes by dividing by γ_0, we get linear equations for the autocorrelations. Specifically, we get for $1 \leq k \leq p$:

$$\gamma(k) = \phi_1\gamma(k-1) + \cdots + \phi_p\gamma(k-p),$$

$$\rho(k) = \phi_1\rho(k-1) + \cdots + \phi_p\rho(k-p).$$

Yule Walker Equations: 2

- If we let $\boldsymbol{\rho} = (\rho(1), \cdots, \rho(p))^T$ and $\mathbf{R}$ be a $p \times p$ matrix whose ij element is $\rho(i-j)$, then

$$\boldsymbol{\rho} = \mathbf{R}\phi.$$

- We want to use these equations, plugging in estimates of $\boldsymbol{\rho}$ to generate an estimate of ϕ.
- Replacing $\rho(k)$ by $\hat{\rho}(k) = r_k,\ 1 \leq k \leq p$ (of course $\rho(0) = \hat{\rho}(0) = 1$) we have the YW estimates for ϕ:

$$\hat{\phi} = \mathbf{R}^{-1}\mathbf{r}.$$

Yule Walker Equations: 3

- If appropriate, one would prefer to use MLE methods for parameter estimation. However, as we will see, the MLEs often must be solved numerically by a search algorithm. This sort of algorithm requires a starting value, and the estimates from the YW equation approach can often provide a good starting value.
- We will soon turn to forecasting/prediction. For this problem, we have data, from which we can fit a model and estimate the model parameters, then we will use that model to forecast future values.

Yule Walker Equations: 4

- For example, if one fit an AR(P) model to data $Y_1, \cdots, Y_t$ and then wished to predict a future value, say Y_{t+k}, the minimum mean square error predictor would be given by $E(Y_{t+k}|\mathcal{F}_t)$. This will often be given by extrapolating the model from time t to time $t+k$ using the estimated coefficients. This will be demonstrated later.
- If one is going to engage in forecasting on a daily (or even shorter) basis, updating the data each time, it would be computationally much more efficient to have a recursive or on-line method to take the old results and update them with the new data.

Yule Walker Equations: 5

- As a simple example of this, consider a stream of numbers, $X_1, X_2, \cdots$. Suppose we want to compute the mean as each new number arrives, so we have the running mean, $\bar{X}_n = \sum_{i=1}^{n} X_i/n$. Each time, rather than going back to the beginning and adding all the numbers up (having to store them all), one can note that

$$\bar{X}_n = \bar{X}_{n-1} + \frac{X_n - \bar{X}_{n-1}}{n}.$$

  Thus, one only needs to know the current value of $n-1$ and $\bar{x}_{n-1}$ and the new number, x_n in order to compute $\bar{x}_n$.
- There is a similar on-line algorithm to update the sample variance knowing only its current value, the sample size and the new value.

Yule Walker Equations: 6

- There is a recursive algorithm of this sort for the solution to the YW equations given by the Durbin Levinson algorithm. We do not present it here, but it is of practical use, especially if one has a current AR(p) model estimated using YW, adds new data and updates the estimated model.
- As we saw in estimating the parameters from a MA(2) model, the problem is far more difficult and must be done numerically. This, of course, carries over to ARMA models which have a MA component. In such cases, the estimation is done using likelihood methods, and standard packages based this on iid white noise errors.

Likelihood Methods: 1

- We begin with the simplest model, AR(1) with a single parameter with iid Normal WN$(0, \sigma_\epsilon^2)$ with three unknown parameters, $\mu, \phi_1, \sigma_\epsilon^2$.
- Suppose the AR(1) data is $(Y_1, Y_2, \cdots, Y_n)$.
- We must make an assumption concerning the distribution of Y_1, the first data point that initiates the process. We might assume that it has the stationary distribution, $N(\mu, \sigma_\epsilon^2/(1-\phi_1^2))$, or that it is unknown.
- Conditional on the starting value, Y_1, the sequence is a Markov process as the conditional distribution of $Y_{i+1}|\mathcal{F}_i$ has a $N(\mu + \phi_1(Y_i - \mu), \sigma_\epsilon^2)$ distribution. Consequently, we can easily write out the likelihood function for the data:

Likelihood Methods: 3

The MLE for σ_ϵ^2 is found by the usual methods and is given by

$$\hat{\sigma}_\epsilon^2 = \frac{1}{n-1}\sum_{i=2}^{n}(Y_i - (\hat{\mu} + \hat{\phi}_1(Y_{i-1} - \hat{\mu}))^2,$$

the residuals from comparing the data with the fitted values for the data using the model.

- The extension of likelihood methods for AR(1) to AR(p) processes is in some sense straightforward, but it will again rely on numerical solutions.
- For AR(p) models and a data set $(Y_1, \cdots, Y_n)$, the model is no longer a Markov process.

Likelihood Methods: 4

- The next value in the time series, Y_{n+1} will depend upon $Y_n, Y_{n-1}, \cdots, Y_{n+1-p}$, but not on any data further in the past. Also there will be an initial part of the data set, $Y_1, \cdots, Y_p$ for which we must determine their joint distribution. One obvious choice is the p-dimensional stationary distribution of the AR(p) process.
- This again calls for numerical methods to find the maximum likelihood, and it is common to use conditional least squares. Nevertheless, the MLE of σ_ϵ^2 is still given by the sum of squared residuals divided by the appropriate denominator.
- Having given this description of the MLE approaches used for the simple AR(1) and AR(p) models, it should be clear that using likelihood methods for MA(1) and ARMA(p,q) models will be more complicated and will certainly require numerical methods.

Statsmodels: 1

- Comment on "statsmodels.tsa.arima_process.arma_generate_sample" There is a parameter in this function, "distrvs" which generates the random variables that drive the process. The default is "np.random.randn" which gives iid N(0,1) errors. This can be changed to other distributions including changing the variance if desired.
- for "statsmodels.tsa.arima_model.ARIMA.fit" there is a parameter called "method" which allow for two choices, "mle" which maximizes the full likelihood including the initial distributions which will be taken from the stationary distribution, or "css" the conditional sum of squares where the initial factors in the likelihood are omitted.

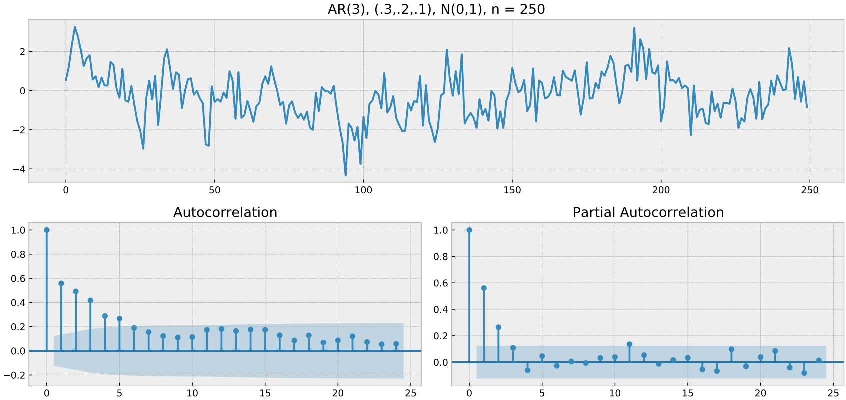
Statsmodels: 2

```
from statsmodels.tsa.arima_process import ArmaProcess
from statsmodels.tsa.arima_model import ARMA
from statsmodels.tsa.stattools import arma_order_select_ic
arparams = np.array([.30, .20, .10])
ar = np.r_[1, -arparams]
ar_process = ArmaProcess(ar)
ar_process.isstationary
yes
y = ar_process.generate_sample(250)
tsplot(y)
plt.show()
res = arma_order_select_ic(y,ic=['aic','bic'],trend = 'nc')
res.aic_min_order
[2,0]
res.bic_min_order
[2,0]
```

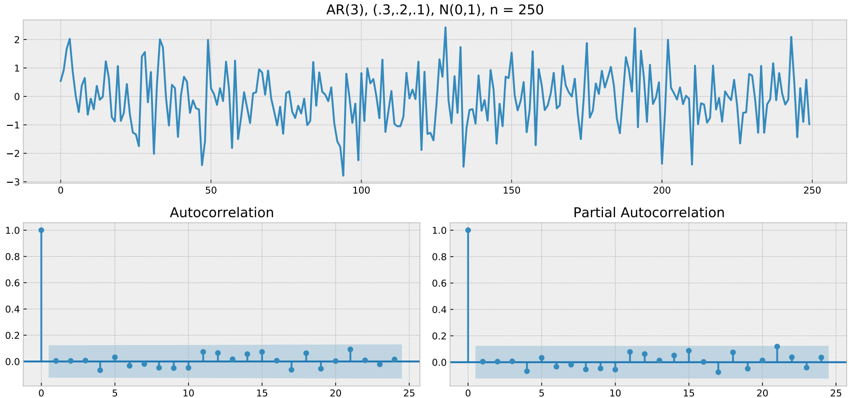
Statsmodels: 3

```
model1 = ARMA(y, (3, 0)).fit(trend='nc', disp=0)
print(model1.params)
tsplot(model1.resid)
plt.show()
model2 = ARMA(y, (2,0)).fit(trend = 'nc', disp = 0)
print(model2.params)
tsplot(model2.resid)
plt.show()
model3 = ARMA(y, (1,1)).fit(trend = 'nc', disp = 0)
print(model3.params)
tsplot(model3.resid)
plt.show()
```

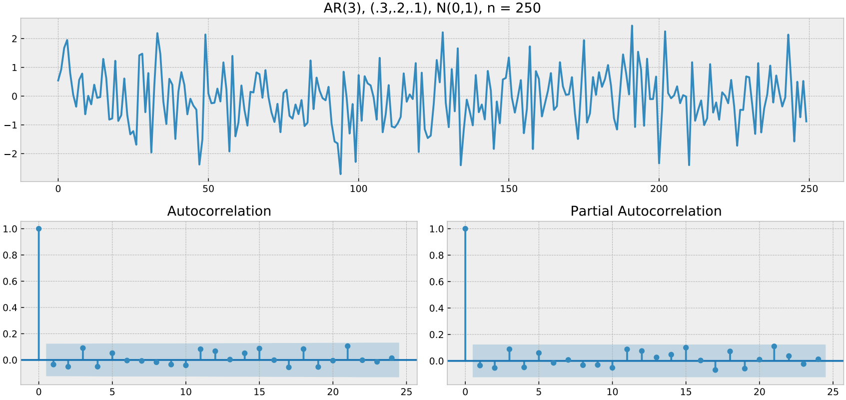
AR(3), (.3,.2,.1), N(0,1), n = 250



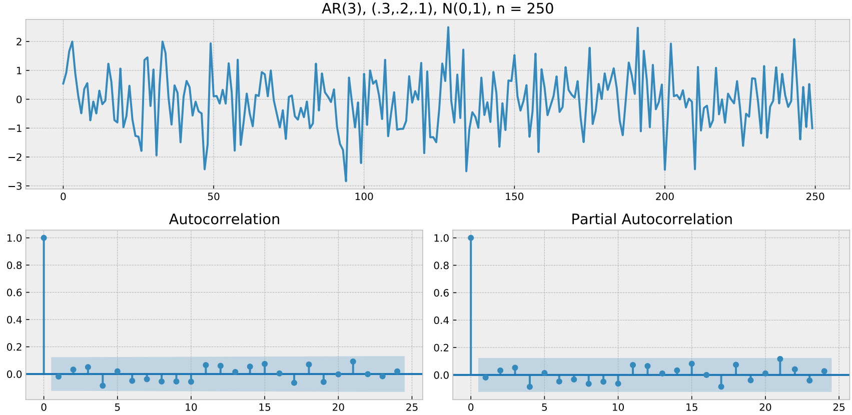
AR(3), (.3,.2,.1), Residuals (3,0),(.38,.22,.12)



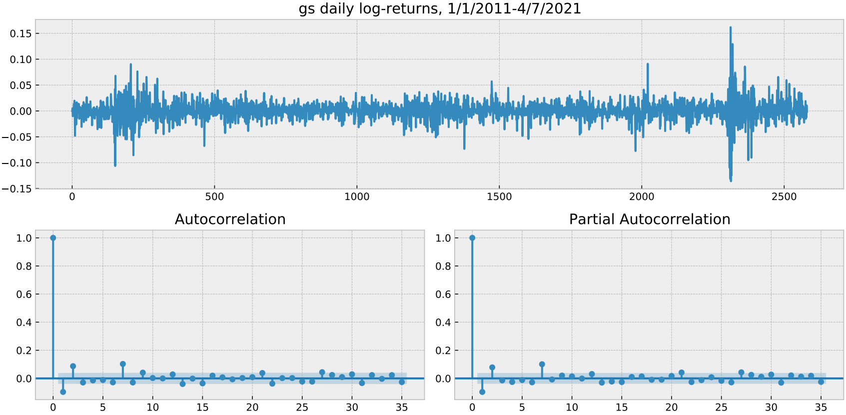
AR(3), (.3,.2,.1), Residuals (2,0), (.42,.27)



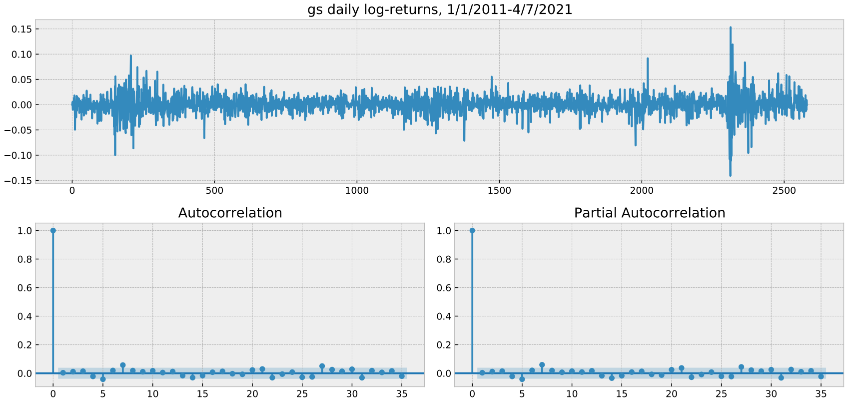
AR(3), (.3,.2,.1), Residuals (1,1),(.86,-.45)



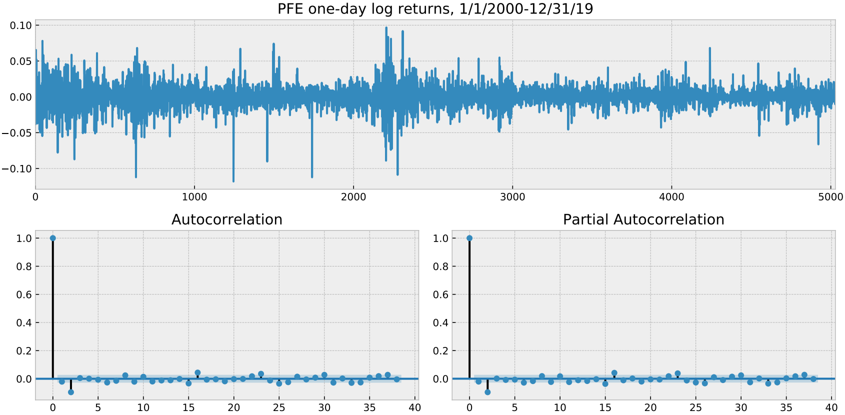
GS daily log-returns 1/1/2011-4/7/2021



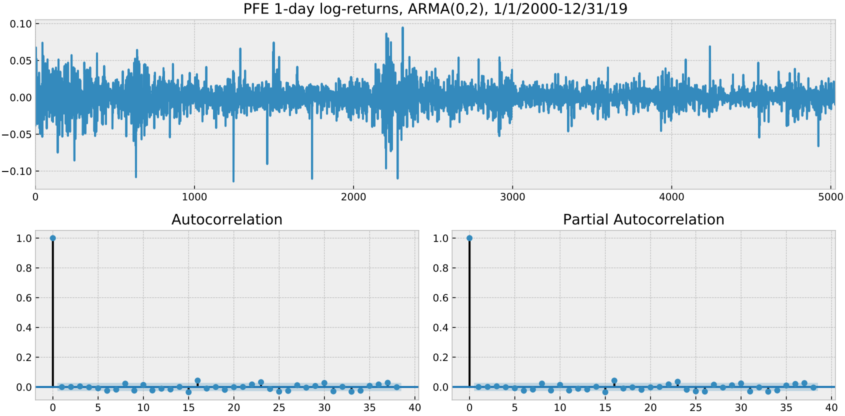
GS daily log-returns residuals from (2,2)



Pfizer (pfe) 1-day log returns, 1/1/2000-12/31/2019



PFE 1-day residuals fit with ARMA(0,2), 1/1/2000-12/31/2019



Forecasting: 1

- We begin with a basic result about prediction. In the context of time series, we have data, $\mathbf{X} = (X_1, X_2, \cdots, X_n)$, and we wish to predict a future value of the series, for example X_{n+k}.
- The prediction must be made using only the available information. We symbolize that knowledge by $\mathcal{F} = \mathcal{F}(\mathbf{X})$.
- The goal is to find the best predictor that is a function of $\mathbf{X}$, that is the function must be measurable with respect to $\mathcal{F}$.
- Our criterion for "best" is the predictor that minimizes the mean square error, $E((Y - predictor)^2)$.
- The answer is $E(Y|\mathcal{F})$.

Forecasting: 2

- Suppose we consider any predictor of Y based on $g(\mathbf{X})$ and we consider its mean square error, $E((Y-g(\mathbf{X}))^2)$.

$$
\begin{aligned}
E((Y-g(\mathbf{X}))^2) &= E((Y-E(Y|\mathcal{F})+E(Y|\mathcal{F})-g(\mathbf{X}))^2) \\
&= E((Y-E(Y|\mathcal{F}))^2)+E(E(Y|\mathcal{F})-g(\mathbf{X}))^2 \\
&= +2E((Y-E(Y|\mathcal{F}))(E(Y|\mathcal{F})-g(\mathbf{X}))) \\
&= E((Y-E(Y|\mathcal{F}))^2)+E(E(Y|\mathcal{F})-g(\mathbf{X})^2) \\
&\geq E((Y-E(Y|\mathcal{F}))^2),
\end{aligned}
$$

where the cross term is 0 because $E(Y-E(Y|\mathcal{F}))=0$, and $E((Y-E(Y|\mathcal{F}))(E(Y|\mathcal{F})-g(\mathbf{X})))=0$ by conditioning the unconditional expectation on $\mathcal{F}$.

Forecasting: 2.5

- Letting $f = E(Y|\mathcal{F})$, an $\mathcal{F}$-measurable function, for the cross term we want to show $E((Y-f)(f-g(\mathbf{X}))) = 0$.

$$\begin{aligned} E((Y-f)(f-g(\mathbf{X}))) &= E(E((Y-f)(f-g(\mathbf{X}))|\mathcal{F}) \\ &= E((f-g(\mathbf{X}))E(Y-f|\mathcal{F})) \\ &= E((f-g(\mathbf{X}))(f-f|)) \\ &= 0 \end{aligned}$$

$$E((Y-g(\mathbf{X}))^2) \geq E((Y-E(Y|\mathcal{F}))^{\in}),$$

for all functions g, that is to say all estimators that are a function of the data $\mathbf{X}$.

- Thus $E(Y|\mathcal{F})$ is the minimum mean square error estimator of Y.

Forecasting: 3

- Given we know the form of the optimal forecasting/prediction function, in full generality it may not be easy to determine.
- We begin with the simplest time series model, AR(1):

$$Y_t = \mu + \phi_1(Y_{t-1} - \mu) + \epsilon_t.$$

- Suppose we have data $(Y_1, \cdots, Y_t)$ and we wish to predict Y_{t+1}, a one-step-ahead forecast.
- We usually assume that the noise process, ϵ_t, is $\mathrm{WN}(0, \sigma_\epsilon^2)$. If we specialize to the case where it is an independent sequence or an MDS, the $E(\epsilon_{t+1}|\mathcal{F}_t) = 0$. So our best prediction of the noise term, ϵ_{t+1}, given the present information is 0.

Forecasting: 4

- The AR(1) model gives $Y_{t+1} = \mu + \phi_1(Y_t - \mu) + \epsilon_{t+1}$, so the best linear forecast will be

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\phi}_1(Y_t - \hat{\mu}) + 0,$$

- Extending to forecasts for times further in the future is straightforward; however, we note that the forecast must be based only on the current dataset, $(Y_1, \cdots, Y_t)$, only.

Forecasting: 5

- To forecast Y_{t+2}, the same idea can be used, except we need to forecast both Y_{t+1} and Y_{t+2}:

$$\begin{aligned} \hat{Y}_{t+2} &= \hat{\mu} + \hat{\phi}_1(\hat{Y}_{t+1} - \mu) \\ &= \hat{\mu} + \hat{\phi}_1(\hat{\phi}_1(Y_t - \hat{\mu})) \\ &= \hat{\mu} + \hat{\phi}_1^2(Y_t - \hat{\mu}). \end{aligned}$$

In general, the k-step-ahead predictor for an AR(1) sequence is

$$\hat{Y}_{t+k} = \hat{\mu} + \hat{\phi}_1^k(Y_t - \hat{\mu}).$$

Forecasting: 6

- While we don't write out the derivation here, all the above ideas extend to AR(p) processes, except the model now contains p parameters whose estimates are used in the forecast.
- A new aspect arises when forecasting a MA(q) time series, therefore also in forecasting an ARMA(p,q) or ARIMA(p,d,q) time series model.
- To see the issue, consider a MA(1) model:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}.$$

Forecasting: 7

- If we wish to forecast one-step-ahead, Y_{t+1}, then we note that $Y_{t+1} = \mu + \epsilon_{t+1} + \theta_1 \epsilon_t$ and the forecast, $\hat{Y}_{t+1}$ must be based only on $\mathcal{F}_t$.
- Clearly, our best estimate of ϵ_{t+1} is 0, and we will obtain estimates of μ and θ_1 and plug them into the equation. But what do we do with ϵ_t?
- We replace ϵ_t by the residual, $\hat{\epsilon}_t$. Thus the forecast of Y_{t+1} is

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_t.$$

Forecasting: 8

- For a MA(1) model, the autocorrelation is 0 for lags greater than 1. To forecast Y_{t+2} we note that the model gives
$$Y_{t+2} = \mu + \theta_1 \epsilon_{t+1} + \epsilon_{t+2}.$$
- Our best estimates of the two error terms, ϵ_{t+1} and ϵ_{t+2} is 0. This gives
$$\hat{Y}_{t+k} = \hat{\mu},\ k \geq 2.$$

Forecasting: 9

In summary:

- We have a weakly stationary time series, $\{X_t\}$, we have observations from that time series, $(X_1, X_2, \ldots, X_n)$, and we wish to predict a future value of the series, say X_{n+k}, a k-step-ahead predictor, $\hat{X}_{n+k}$.
- The criterion we are trying optimize is minimum mean square error, $E((X_{n+k} - \hat{X}_{n+k})^2)$.
- We know that the optimal estimator is $E(X_{t+k}|\mathcal{F}_n)$ to minimize mean square error.
- In practice, we use the optimum linear forecast, i.e.

$$\hat{X}_{n+k} = a_0 + a_1 X_1 + \cdots + a_n X_n.$$

- As n changes, the coefficients may change, so we actually need to represent this forecast equation as

$$\hat{X}_{n+k} = a_{0n} + a_{1n} X_1 + \cdots + a_{nn} X_n.$$